

```

#include <stdio.h>
#define MAX 20
int a[MAX], n;
void display()
{
    int i;
    printf("\nCurrent array: ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    printf("\n");
}

void insertion(int x, int p)
{
    int i;

    if(n == MAX)
    {
        printf("Array is full.\n");
        return;
    }

    if(p < 1 || p > n + 1)
    {
        printf("Invalid position.\n");
        return;
    }

    for(i = n; i >= p; i--)
    {
        a[i] = a[i - 1];
    }

    a[p - 1] = x;
    n++;
}

void deletion(int p)
{
    int i;

    if(p < 1 || p > n)
    {
        printf("Invalid position.\n");
        return;
    }

    for(i = p - 1; i < n - 1; i++)
    {
        a[i] = a[i + 1];
    }

    n--;
    printf("Element deleted successfully.\n");
}

int search(int x)
{
    int i;

    for(i = 0; i < n; i++)
    {

```

```

        if(a[i] == x)
            return i + 1;
    }

    return -1;
}

int main()
{
    int i, ch, x, p;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    if(n < 0 || n > MAX)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    display();

    do
    {
        printf("\n----- MENU -----");
        printf("\n1. Insert");
        printf("\n2. Delete");
        printf("\n3. Search");
        printf("\n4. Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &ch);

        switch(ch)
        {
            case 1:
                printf("Enter element: ");
                scanf("%d", &x);
                printf("Enter position: ");
                scanf("%d", &p);
                insertion(x, p);
                display();
                break;

            case 2:
                printf("Enter position to delete: ");
                scanf("%d", &p);
                deletion(p);
                display();
                break;

            case 3:
                printf("Enter element to search: ");
                scanf("%d", &x);

                p = search(x);

                if(p != -1)
                    printf("Element found at position %d\n", p);
                else

```

```
        printf("Element not found.\n");
        break;

    case 4:
        printf("Program terminated.\n");
        break;

    default:
        printf("Invalid choice.\n");
    }

} while(ch != 4);

return 0;

}
```

Enter number of elements: 5

Enter array elements:

11 12 13 14 16

Current array: 11 12 13 14 16

----- MENU -----

1. Insert

2. Delete

3. Search

4. Exit

Enter your choice: 1

Enter element: 2

Enter position: 2

Current array: 11 2 12 13 14 16

----- MENU -----

1. Insert

2. Delete

3. Search

4. Exit

Enter your choice: 2

Enter position to delete: 3

Element deleted successfully.

Current array: 11 2 13 14 16

----- MENU -----

1. Insert
2. Delete
3. Search
4. Exit

Enter your choice: 3

Enter element to search: 16

Element found at position 5

----- MENU -----

1. Insert
2. Delete
3. Search
4. Exit

Enter your choice: 4

Program terminated.

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Array Program (insert, delete, search)

```
#include <stdio.h>
```

```
#define MAX 20
```

```
int a[MAX], n;
```

```
void display ()
```

```
{  
    int i;
```

```
    printf ("In current array : ");
```

```
    for (i=0; i<n; i++)
```

```
    {  
        printf ("%d ", a[i]);
```

```
    }
```

```
    printf ("\n");
```

```
}
```

```
void _insertion (int x, int p)
```

```
{  
    int i;
```

```
if (n == max)
```

```
{  
    printf ("Array is full.\n");  
    return;  
}
```

```
if (p < 1 || p > n + 1)
```

```
{  
    printf ("Invalid position.\n");  
    return;  
}
```

```
}
```

```
for (i = n; i >= p; i--)
```

```
{  
    a[i] = a[i-1];
```

```
}
```

```
a[p-1] = x;
```

```
n++;
```

```
}
```

```
void deletion (int p)
```

```
{  
    int i;
```

```
    if (p < 1 || p > n)
```

```
{  
    printf ("Invalid position.\n");  
    return;  
}
```

```
}
```

```
for (i = p-1; i < n-1; i++)
```

```
{  
    a[i] = a[i+1];
```

```
}
```

```
n--;
```

```
printf ("Elements are deleted successfully.\n");
```

```
}
```

```
int search (int x)
```

```
{
```

```
    int i;
```

```
for (i=0; i<n; i++)
```

```
{ if (a[i] == x)
```

```
    return i+1;
```

```
}
```

```
return -1;
```

```
}
```

```
int main()
```

```
{
```

```
    int i, ch, x, n;
```

```
    printf ("Enter number of elements: ");
```

```
    scanf ("%d", &n);
```

```
    if (n <= 0 || n > MAX)
```

```
    {
```

```
        printf ("Invalid array size.\n");
```

```
        return 0;
```

```
    }
```

```
    printf ("Enter array elements: \n");
```

```
    for (i=0; i<n; i++)
```

```
    {
```

```
        scanf ("%d", &a[i]);
```

```
    }
```

```
    display();
```

```
do
```

```
{
```

```
    printf ("\n 1. Insert");
```

```
    printf ("\n 2. Delete");
```

```
    printf ("\n 3. Search");
```

```
    printf ("\n 4. Exit");
```

```
    printf ("\n Enter your choice: ");
```

```
    scanf ("%d", &ch);
```

switch (ch)

{

case 1:

```
printf ("Enter element: ");  
scanf ("%d", &x);  
printf ("Enter position: ");  
scanf ("%d", &p);  
insertion (x, p);  
display ();  
break;
```

case 2:

```
printf ("Enter position to delete: ");  
scanf ("%d", &p);  
deletion (p);  
display ();  
break;
```

case 3:

```
printf ("Enter element to search: ");  
scanf ("%d", &x);  
p = search (x);
```

~~if (p != -1)~~

~~printf ("Element found at position %d\n", p);~~

~~else~~

~~printf ("Element not found. \n");~~

~~break;~~

case 4:

printf ("program terminated. \n");

break;

default;

printf("Invalid choice.\n");

}

while (ch != 4);

return 0;

}

Output

Enter number of elements:

Enter Array elements:

11 12 13 14 16

----- MENU -----

1. Insert

2. Delete

3. Search

4. Exit

Enter your choice: 1

Enter element: 2

Enter position: 2

Current Array: 11 12 2 13 14 16

Enter your choice: 2

Enter position to delete: 3

Element deleted successfully

Current Array: 11 2 13 14 16

Enter your choice: 3

Enter element to search: 16

Element found at position: 5

Enter your choice: 4

Program terminated.

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